

9. $\begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} =$

1) 3

2) 2

3) -3

4) -2

10. Let P, Q, R be the mid points of AB, BC, CA of ΔABC . If the area of ΔABC is 20 sq.units, then the area of ΔPQR is _____sq. units

1) 100

2) 5

3) 10

4) 3

JEE MAIN & ADVANCED**LEVEL - 1****SINGLE CORRECT CHOICE TYPE:**

1. If $\begin{vmatrix} 2a & 5 \\ 6 & 3 \end{vmatrix} = 0$, then a =

1) 3

2) 0

3) 5

4) -5

2. If $A = \begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix}$ and $B = \begin{vmatrix} 1 & 1 \\ 2 & 2 \end{vmatrix}$, then the value of 'AB' is

1) 5

2) 6

3) 0

4) -6

3. $\begin{vmatrix} 1 & \sin \theta \\ \sin \theta & 1 \end{vmatrix} =$

1) $\sin^2 \theta$ 2) $\cos^2 \theta$ 3) $-\cos^2 \theta$ 4) $-\sin^2 \theta$

4. $\begin{vmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{vmatrix} =$

1) 0

2) 18

3) 95

4) 20

5. The area of the triangle formed by the points $A(0,0), B(x_2, y_2), C(x_3, y_3)$ is

1) $\frac{1}{2}(x_2 y_3 - x_3 y_2)$

2) $\frac{1}{2}|\sum x_1(y_3 - y_2)|$

3) $\frac{1}{2}|\sum y_1(x_1 - x_2)|$

4) $\frac{1}{2}|\sum y_2(x_2 - x_3)|$

6. The area of quadrilateral whose vertices, taken in order are $(-4, -2), (-3, -5), (3, -2), (2, 3)$ is

1) 28 sq.units

2) 26 sq.units

3) 24 sq.units

4) 22 sq.units

MULTI CORRECT CHOICE TYPE:

7. $\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} =$

1) 1

2) -1

3) $\begin{vmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{vmatrix}$

4) 2

8. $\begin{vmatrix} 3421 & 3422 \\ 3423 & 3424 \end{vmatrix} =$

1) -2

2) $\begin{vmatrix} i & 1 \\ 1 & i \end{vmatrix}$

3) 0

4) -1

STATEMENT TYPE:

9. **Statement-I:** The area of the triangle formed by the points (1,2), (3,4) and (5,6) is 10sq.units

Statement-II: If “G” is the centroid of the triangle ABC such that area of

$$\Delta AGB = \frac{1}{3}(\text{Area of } \Delta ABC).$$

- 1) Both statements I and II are true
 2) Both statements I and II are false
 3) Statement – I is true but statement – II is false
 4) Statement – I is false but statement – II is true

COMPREHENSION TYPE:

The area of the triangle formed by the points A(x₁, y₁), B(x₂, y₂) and C(x₃, y₃) is

$$\frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| \text{ sq. units.}$$

10. If the centroid of a triangle is (1,4) and the two of its vertices are (4,-3) and (-9,7), then the area of the triangle is _____ sq. units
- 1) $\frac{183}{2}$ 2) 183 3) 336 4) 122
11. If D(1,2), E(3,4) and F(5,7) are the mid point of the sides of a triangle ΔABC , then the area of the ΔABC is _____ sq. units.
- 1) 4 2) 1 3) 2 4) 3
12. If the points (t, 2t), (-2, 6), (3, 1) are collinear, then the value of ‘t’ is
- 1) $\frac{3}{4}$ 2) $\frac{4}{3}$ 3) 3 4) 4

MATRIX MATCH TYPE:13. **Column-I**

a) $\begin{vmatrix} a & ib \\ ib & a \end{vmatrix} =$

b) $\begin{vmatrix} a & -ib \\ ib & a \end{vmatrix} =$

c) $\begin{vmatrix} a & b \\ b & a \end{vmatrix} =$

d) $\begin{vmatrix} a & b \\ -b & a \end{vmatrix} =$

Column-II

p) $a^2 - b^2$

q) $(a+b)^2 - 2ab$

r) $a^2 + b^2$

s) $(a+b)(a-b)$

t) $2ab$

INTEGER ANSWER TYPE:

14. The area of the right angled triangle whose vertices are $(2,-2), (-2,1)$ and $(5,2)$ is $\frac{25}{k}$ sq.units, then $k =$ _____.

JEE MAIN & ADVANCED**LEVEL - 2****SINGLE CORRECT CHOICE TYPE:**

15. $\begin{vmatrix} 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{vmatrix} =$

1) 1

2) 2

3) 4

4) 9

16. The area of the triangle formed by the points $(a, b+c), (b, c+a), (c, a+b)$ is

1) abc sq. units

2) $2ab$ sq. units

3) $3abc$ sq. units

4) 0 sq. units

17. The real part of $\begin{vmatrix} \cos \alpha + i \sin \alpha & \cos \beta + i \sin \beta \\ \sin \beta + i \cos \beta & \sin \alpha + i \cos \alpha \end{vmatrix}$ is

1) 1

2) -1

3) 0

4) 2

MULTI CORRECT CHOICE TYPE:

18. If the points $(a,0), (0,b), (1,1)$ are collinear, then an incorrect statement is

1) $\frac{1}{a^2} + \frac{1}{b^2} = 1$

2) $\frac{1}{a^2} - \frac{1}{b^2} = 1$

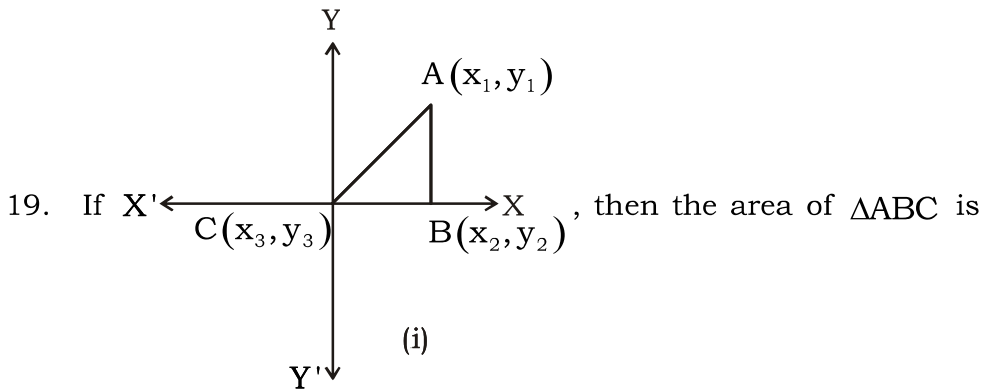
3) $\frac{1}{a} + \frac{1}{b} = 1$

4) $\frac{1}{a} - \frac{1}{b} = 1$

COMPREHENSION TYPE:

The area of the triangle formed by the points $A(x_1, y_1), B(x_2, y_2)$ and $C(x_3, y_3)$ is

$$\frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| \text{ sq. units.}$$

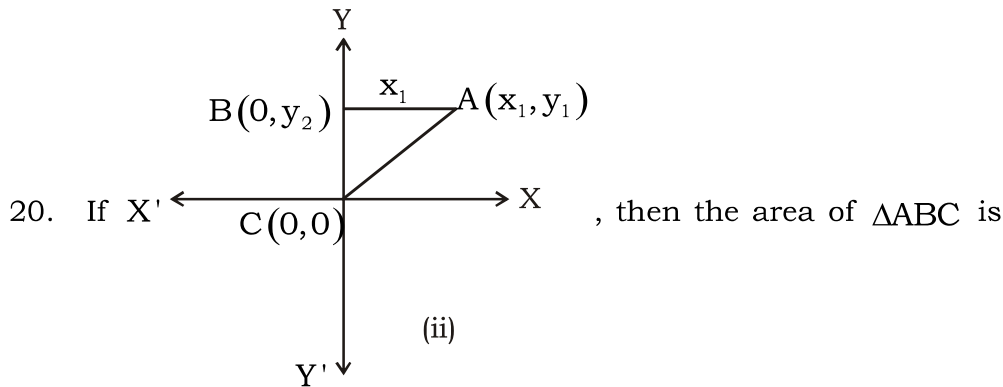


1) $\frac{1}{2} |x_1(y_2 - y_1) - x_3(y_2 - y_1)| \text{ sq. units}$

2) $\frac{1}{2} |x_1 y_2| \text{ sq. units}$

3) $\frac{1}{2} |x_1(y_2 - y_3) - x_2(y_2 - y_3)| \text{ sq. units}$

4) $\frac{1}{2} |x_2 y_1| \text{ sq. units}$

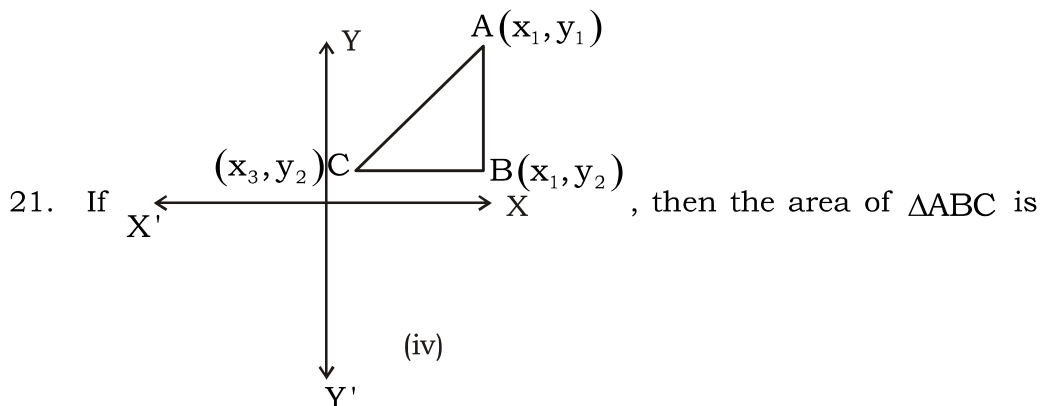


1) $\frac{1}{2} |x_1(y_2 - y_1) - x_3(y_2 - y_1)| \text{ sq. units}$

2) $\frac{1}{2} |x_1 y_2| \text{ sq. units}$

3) $\frac{1}{2} |x_1(y_2 - y_3) - x_2(y_2 - y_3)| \text{ sq. units}$

4) $\frac{1}{2} |x_2 y_1| \text{ sq. units}$



1) $\frac{1}{2} |x_1(y_2 - y_1) - x_3(y_2 - y_1)| \text{ sq. units}$

2) $\frac{1}{2} |x_1 y_2| \text{ sq. units}$

3) $\frac{1}{2} |x_1(y_2 - y_3) - x_2(y_2 - y_3)| \text{ sq. units}$

4) $\frac{1}{2} |x_2 y_1| \text{ sq. units}$

JEE MAIN & ADVANCED**LEVEL - 3****SINGLE CORRECT CHOICE TYPE:**

22. If $\begin{vmatrix} x & 3 & 6 \\ 3 & 6 & x \\ 6 & x & 3 \end{vmatrix} = \begin{vmatrix} 2 & x & 7 \\ x & 7 & 2 \\ 7 & 2 & x \end{vmatrix} = 0$, then $x =$

1) -9

2) -1

3) 2

4) -2

23. The area of the triangle formed by $(a+3, a-2)$, $(a-4, a+5)$ and (a, a) is

1) 0 sq. units

2) a sq. units

3) $\frac{7}{2}$ sq. units4) a^2 sq. units**MATRIX MATCH TYPE:**

24. If $(p+1, 1)$, $(2p+1, 3)$ and $(2p+2, 2p)$ are the vertices of a triangle then match the following.

Column-I**Value of p**a) $p = 0$ b) $p = \pm 1$ c) $p = 3$ d) $p = -3$ **Column-II****Area of the triangle**p) $7/2$ q) $25/2$ r) $3/2$

s) 1

INTEGER ANSWER TYPE:

25. The area of the triangle formed by $(0,0)$, $(a^{x^2}, 0)$, $(0, a^{6x})$ is $\frac{1}{2a^5}$ sq. units, and $x = -k$, then $k =$ _____.

SEEKING TO DESCRIBE

26. If $\begin{vmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 0 & 0 & \lambda \end{vmatrix} = 70$, then find the value of ' λ '.